Low-Light Image Enhancement Framework

Computer Vision Course Project – Bachelor Level (BSc)

Low-Light Image Enhancement using Classical and Lightweight Learning Methods

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1. Introduction

Images captured in low-light conditions often suffer from poor visibility, low contrast, noise amplification, and loss of structural details. These degradations significantly reduce the effectiveness of image analysis systems and negatively impact both human perception and automated vision algorithms.

The goal of this project is to provide undergraduate students with a **deep yet manageable learning experience** in low-light image enhancement. Students will explore classical image processing techniques alongside a lightweight learning-based method, analyze their strengths and limitations, and quantitatively evaluate enhancement quality using well-established metrics.

This project emphasizes **analysis, understanding, and comparison**, rather than large-scale training or industrial-level systems.

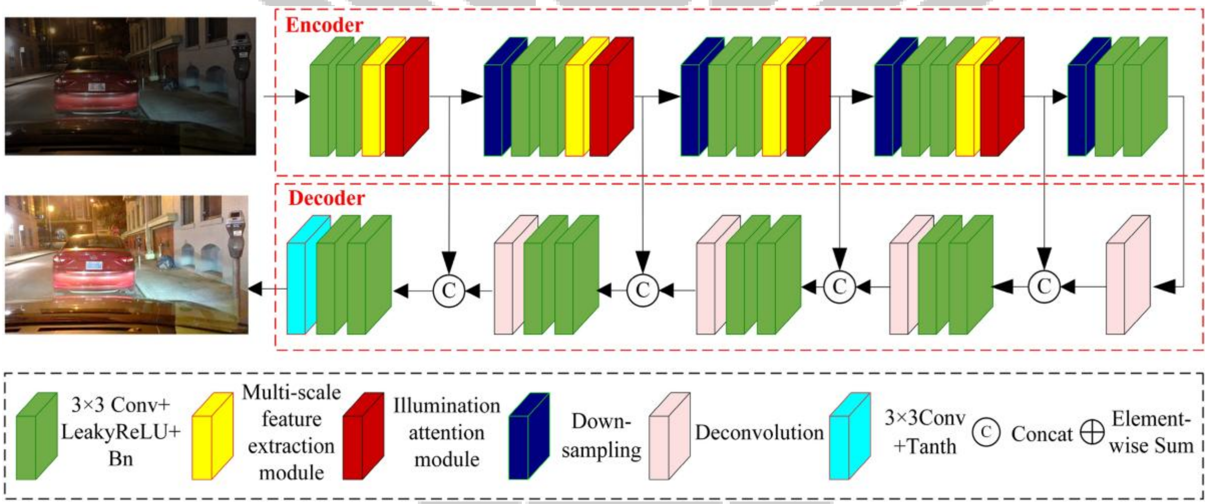

Figure 1: Illustration of enhancing a low-light image to improve visibility and quality.

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2. Dataset

This project uses a **subset of the LOL-v2 (Low-Light) Dataset – Real Part Only**.

Dataset Characteristics

- Real-world low-light images paired with normal-light reference images
- Indoor and outdoor scenes
- Authentic illumination challenges

Usage Constraints

- Students must use **80–120 paired images only**
- Images must be resized to a fixed resolution (e.g., 256×256)
- The dataset may be randomly split into training and testing subsets where required

Dataset link will be provided by the teaching assistants.

Dataset Examples

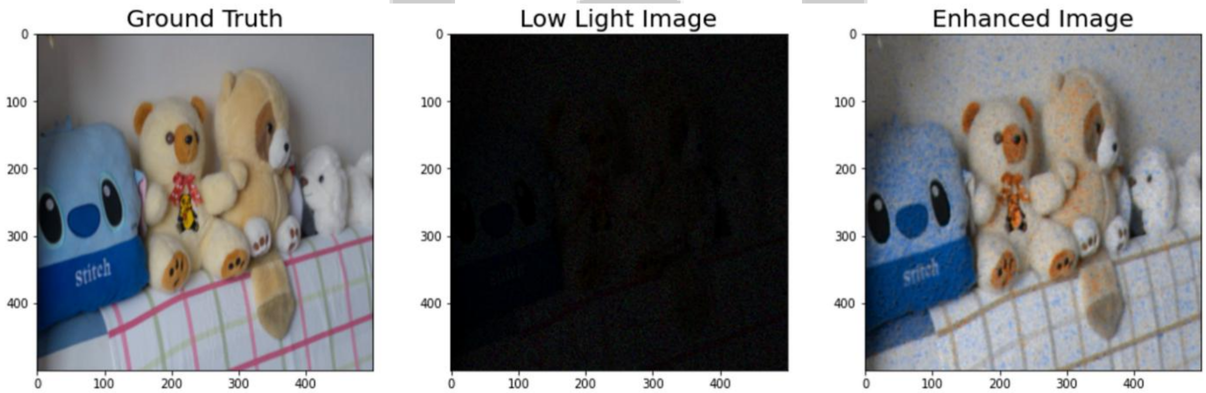

Figure 2: Examples from LOL-v2 dataset: (a) Low-light image, (b) Normal-light reference image.

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3. Preprocessing

Preprocessing ensures consistency and stability across all experiments.

Required preprocessing steps: 1. **Image Resizing:** Standardize all images to the same resolution 2. **Normalization:** Scale pixel values to [0, 1] 3. **Optional Augmentation:** Horizontal flipping or small rotations

All preprocessing choices must be clearly justified in the report.

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4. Project Structure and Phases

The project consists of **three structured phases**:

1. Image Quality Analysis and Classification
2. Image Enhancement
3. Quantitative and Qualitative Evaluation

4.1 Phase 1 – Image Quality Analysis and Classification

Objective

Analyze the statistical differences between low-light and normal-light images and train a lightweight classifier using handcrafted features.

Tasks

Feature Extraction

Students must extract the following features from grayscale images: - Mean Intensity - Standard Deviation (Contrast) - Entropy - Histogram Skewness

Classification

- Train a **binary classifier** (Low-Light vs Normal-Light)
- Allowed models:
  - Logistic Regression
  - Support Vector Machine (SVM)

Deep learning models are **not allowed** in this phase.

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix visualization

Deliverables (Phase 1)

- Feature extraction code
- Trained classifier
- Evaluation table and confusion matrix

4.2 Phase 2 – Image Enhancement

This phase forms the **core of the project**.

Part A – Classical Enhancement Methods (Mandatory)

Students must implement and analyze **at least three** of the following methods:

1. Histogram Equalization
2. CLAHE (Contrast Limited Adaptive Histogram Equalization)
3. Gamma Correction (with parameter sensitivity analysis)

4. Single-Scale Retinex (SSR)

Each method must be implemented using OpenCV or NumPy.

Part B – Lightweight Learning-Based Enhancement (Mandatory)

Students must implement **one lightweight deep learning model**:

Autoencoder-Based Enhancement

- Simple convolutional autoencoder
- 3–4 convolutional layers
- MSE loss only
- No GANs, no perceptual loss
- Training limited to 20–30 epochs

The model learns to map low-light images to their normal-light counterparts.

4.3 Phase 3 – Evaluation and Analysis

Quantitative Metrics

To evaluate the performance of your enhancement framework, you will use two complementary metrics that assess different aspects of image quality. These metrics quantify how well your enhanced images match the normal-light reference images.

1.Peak Signal-to-Noise Ratio (PSNR)

Definition

PSNR is a widely-used metric that measures the ratio between the maximum possible signal power and the power of corrupting noise. It provides an objective measure of image quality based on pixel-wise differences.

Formula

$$\text{PSNR} = 10 \cdot \log_{10}\left(\frac{MAX_I^2}{MSE}\right)$$

where: - MAX_I = maximum possible pixel value (255 for 8-bit images) - MSE = Mean Squared Error between enhanced and reference images

$$MSE = \frac{1}{M \times N} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} [I(i,j) - K(i,j)]^2$$

where: - $I(i,j)$ = enhanced image pixel value at position (i,j) - $K(i,j)$ = reference image pixel value at position (i,j) - $M \times N$ = image dimensions

Interpretation

- **Higher PSNR = Better quality**
- Measured in decibels (dB)
- Typical ranges:
  - **< 20 dB**: Poor quality
  - **20-25 dB**: Acceptable quality
  - **25-30 dB**: Good quality
  - **> 30 dB**: Excellent quality
  - **> 40 dB**: Nearly identical to reference

Advantages

- Simple to calculate
- Widely recognized standard
- Easy to interpret

Limitations

- Does not always correlate well with human perception
- Sensitive to pixel misalignments
- Treats all types of errors equally

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2. Structural Similarity Index (SSIM)

Definition

SSIM is a perceptual metric that assesses the structural similarity between two images. Unlike PSNR, SSIM considers three components that align with human visual perception: **luminance**, **contrast**, and **structure**.

Formula

$$SSIM(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

where: - x, y = enhanced and reference images (or local windows) - μ_x, μ_y = average pixel values (luminance) - σ_x^2, σ_y^2 = variances (contrast) - σ_{xy} = covariance (structural correlation) - C_1, C_2 = small constants to stabilize division -

$C_1 = (K_1 \cdot L)^2$ - $C_2 = (K_2 \cdot L)^2$ - where L = dynamic range (255 for 8-bit), $K_1 = 0.01$, $K_2 = 0.03$

Three Components

5. **Luminance Comparison**:

$$l(x,y) = \frac{2\mu_x\mu_y + C_1}{\mu_x^2 + \mu_y^2 + C_1}$$

6. **Contrast Comparison**:

$$c(x,y) = \frac{2\sigma_x\sigma_y + C_2}{\sigma_x^2 + \sigma_y^2 + C_2}$$

7. **Structure Comparison**:

$$s(x,y) = \frac{\sigma_{xy} + C_3}{\sigma_x\sigma_y + C_3}$$

Interpretation

- **Range**: -1 to 1 (typically 0 to 1 for similar images)
- **1 = Perfect similarity** (identical images)
- **0 = No similarity**
- Typical ranges:
  - **< 0.6**: Poor quality
  - **0.6-0.8**: Acceptable quality
  - **0.8-0.9**: Good quality
  - **> 0.9**: Excellent quality

Advantages

- Better correlation with human perception than PSNR
- Symmetric: SSIM(x,y) = SSIM(y,x)
- Considers local image structure
- More robust to uniform brightness/contrast changes

Limitations

- More computationally expensive than PSNR
- Requires careful parameter selection
- May not capture all aspects of perceptual quality

Metrics must be reported for: - Original low-light images - Each enhancement method - Autoencoder output

Visualization Requirements

- Bar charts comparing PSNR and SSIM across methods
- Visual comparison: input vs enhanced vs reference images

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5. Analysis and Discussion (Critical Section)

Students must provide a **detailed written analysis** addressing the following questions:

1. Why can higher PSNR values sometimes correspond to worse visual quality?
2. In which scenarios do classical methods outperform the autoencoder?
3. What is the trade-off between brightness enhancement and noise amplification?
4. Is increased brightness always equivalent to better image quality?

This section is heavily weighted in grading.

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6. Deliverables

Each student (or group) must submit:

1. A clean and executable **Jupyter Notebook**
2. A **technical report (8–10 pages)** including:
    - Introduction
    - Methodology
    - Experiments
    - Results
    - Discussion and Conclusion
3. Visual comparison figures

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7. Grading Criteria

Component	Weight
Phase 1 – Analysis & Classification	20%
Phase 2 – Enhancement Methods	40%
Phase 3 – Evaluation & Analysis	30%
Report Quality & Clarity	10%
Total	**100%**

Bonus (Optional)

- Best analysis and insight: +5%
- Best visual results: +5%

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8. Academic Integrity

- Plagiarism or code copying will result in zero credit
- All sources and libraries must be properly cited

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9. Contact Information

For questions and clarifications, contact the teaching assistants:

- **Ali Rezaei** –
- Email: Alirezait95@gmail.com
- Telegram: @AliGeeky_RZ
- **Ehsan Rostami** –
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- Telegram: @e_rostmi

Response time: 24–48 hours

Good luck, and enjoy exploring low-light image enhancement!